🧪 INORGANIC CHEMISTRY — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

📄 Topics in this chapter:

1. Periodic Table — History & Periodic Trends
2. Oxides & Valency
3. Elemental Abundance — Earth & Universe
4. How Many Elements? — Modern Count
5. Element Classification & Diamond
6. First Artificial Element — Technetium

1. Periodic Table — History & Periodic Trends

- UNESCO ne 2019 ko International Year of the Periodic Table ke roop mein celebrate kiya. Ye Mendeleev ki 1869 wali periodic table ke 150 years complete hone ki anniversary thi. [Q1]
- Group mein top se bottom atomic radius generally increase hoti hai, kyunki electron shells badhte hain. Electron affinity ka overall trend generally less favourable/decrease hota hai; isliye "hamesha increase hoti hai" assertion false hai, jabki radius wala reason true hai. [Q2]
- Period mein left se right effective nuclear attraction badhne ke karan atomic radius generally decrease, ionization potential generally increase aur electronegativity generally increase karti hai. Group mein neeche jaane par electron affinity ka simple exam trend decrease maana gaya hai. [Q3]

⚠ EXAM TRAP — Electron Affinity Trend — "Always" se bachna

▸ Electron affinity ka trend completely smooth nahi hota; Be/Mg, N/P, noble gases aur halogens mein notable exceptions/sign-convention issues milte hain. Exam mein "always" word ho to statement ko carefully evaluate karo. [Q2, Q3]

2. Oxides & Valency

- Source ke according Group 3 aur Group 4 ke oxides mein basic aur acidic dono behaviour mil sakta hai, yani amphoteric nature poochha gaya hai. [Q4]
- Common valency mapping: Silicon = 4, Fluorine = 1, Aluminium = 3, Sulphur = 2. Isliye correct match Si-(iv), F-(i), Al-(iii), S-(ii) hai. [Q5]

⚠ EXAM TRAP — Q4 — Group 3/4 Oxides ka broad generalisation

▸ Har Group 3 aur Group 4 oxide ko universally "amphoteric" kehna oversimplification hai. Different oxidation states aur elements ke oxides basic, amphoteric ya acidic character dikha sakte hain; source answer ko exam-context mein yaad rakho. [Q4]

3. Elemental Abundance — Earth & Universe

- Earth's crust mein mass percentage ke hisab se Oxygen sabse abundant element hai — roughly 46.6%. Uske baad Silicon lagbhag 27.7%, phir Aluminium aur Iron aate hain. [Q6, Q7]
- Oxygen ke baad Earth's crust mein Silicon second most abundant element hai. Silicon semiconductor material ke roop mein computer/electronic chips mein extensively use hota hai. [Q8]
- Universe mein Hydrogen sabse abundant chemical element hai; ordinary/baryonic matter ke mass ka roughly three-fourths Hydrogen se aur major remaining share Helium se bana hai. [Q9]

⚠ EXAM TRAP — Q8 — "Earth's Surface" vs "Earth's Crust"

▸ Source wording "Earth's surface" use karti hai, lekin standard abundance fact crust-by-mass ke context mein Oxygen first aur Silicon second hai. Question ko crust/surface wording ke saath confuse mat karo. [Q8]

4. How Many Elements? — Modern Count

- Modern periodic table mein 118 identified/named chemical elements hain. Q10 ke options 300, 250, 200 aur 100 the; inmein exact correct answer nahi tha, isliye source ne question ko disputed (*) mark kiya hai. [Q10]

⚠ EXAM TRAP — Q10 — Correct Modern Count = 118

▸ 118 accepted elements hain. “100” sirf historical options mein closest choice hai; factually exact answer nahi. Aise question mein current scientific count ko priority do. [Q10]

5. Element Classification & Diamond

- Diamond chemically crystalline Carbon ka allotrope hai, isliye ye elemental form hai. Sand mainly SiO_2, marble mainly $CaCO_3$ aur sugar carbon-hydrogen-oxygen containing compound hoti hai. [Q11, Q12]
- Periodic-table elements ko broadly metals, non-metals aur metalloids mein classify kiya jata hai. “Gas” ek physical state of matter hai, element ki fundamental classification category nahi. [Q13]

6. First Artificial Element — Technetium

- Technetium (Tc), atomic number 43, ko first artificially produced element maana jata hai. Carlo Perrier aur Emilio Segrè ne 1937 mein is element ko identify/isolate kiya. [Q14]
- Symbol-based question mein first artificially prepared element = Tc (Technetium). Te = Tellurium, Th = Thorium aur Tl = Thallium is answer ke alternatives nahi hain. [Q15]